Program : **Diploma in Cyber Forensics and Information security**	
Course Code : **6282B**	Course Title: **Data Mining and Warehousing**
Semester : **6**	Credits: **4**
Course Category: **Open elective course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives

- To introduce the concepts of data mining and its applications
- To understand investigation of data using practical data mining tools.
- To introduce Association Rules in mining and need of clustering
- To introduce advanced Data Mining techniques

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge on computer concepts.		Introduction to IT Systems Lab	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the fundamental concepts of data mining and its applications	13	Understanding
CO2	Understand investigation of data for preprocessing using practical data mining tools	15	Understanding Applying
CO3	Understand Association Rules in data mining and need of clustering	16	Understanding Applying
CO4	Understand advanced Data Mining techniques	14	Understanding Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3			2				1
CO4	2						1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Interpret the fundamental concepts of data mining and its applications**		
M1.01	Concepts and applications	2	Understanding
M1.02	Data Mining Stages	2	Understanding
M1.03	Data Mining Models	2	Understanding
M1.04	Data Warehousing and OLAP	3	Understanding
M1.05	Challenges and applications	4	Understanding
Contents: Module 1 Data Mining:- Concepts and Applications, Data Mining Stages, Data Mining Models, Data Warehousing (DWH) and On-Line Analytical Processing (OLAP), Need for Data Warehousing, Challenges, Application of Data Mining Principles, OLTP Vs DWH, Applications of DWH			
CO2	**Understand investigation of data for preprocessing using practical data mining tools**		
M2.01	Need of Data preprocessing	3	Understanding
M2.02	Data Preprocessing Concepts	3	Understanding
M2.03	Data integration and transformation	3	Understanding
M2.04	Data Reduction, Discretization and concept hierarchy.	3	Understanding

M2.05	Classification Models	3	Understanding Applying
	Series Test – I	1	
Contents : Module 2 Data Preprocessing: Data Preprocessing Concepts, Data Cleaning, Data integration and transformation, Data Reduction, Discretization and concept hierarchy. Classification Models: Introduction to Classification and Prediction, Issues regarding classification and prediction,			
CO3	**Understand Association Rules in data mining and need of clustering**		
M3.01	Association Rules Mining	3	Understanding
M3.01	Apriori	2	Understanding
M3.02	FP-Growth	2	Understanding
M3.03	Cluster Analysis	3	Understanding Applying
M3.04	Categorization of clustering methods	3	Understanding Applying
M3.05	Partitioning method	3	Understanding Applying
Contents: Module 3 Association Rules Mining: Concepts, Apriori and FP-Growth Algorithm. Cluster Analysis: Introduction, Concepts, Types of data in cluster analysis, Categorization of clustering methods, Partitioning method: K-Means and K-Medoid Clustering.			
CO4	**Understand advanced Data Mining techniques**		
M4.01	Hierarchical Clustering method	3	Understanding
M4.02	Advanced Data Mining Techniques	4	Understanding
M4.03	Graph mining	3	Understanding Applying
M4.04	Social Network Analysis	4	Understanding Applying
	Series Test – II	1	

Contents : Module 4
Hierarchical Clustering method: BIRCH. Density-Based Clustering –DBSCAN and OPTICS. Advanced Data Mining Techniques: Introduction, Web Mining- Web Content Mining, Web Structure Mining, Web Usage Mining. Text Mining. Graph mining:- Apriori based approach for mining frequent subgraphs. Social Network Analysis:- characteristics of social networks. Link mining:- Tasks and challenges.

Text / References

T/R	Book Title/Author
T1	Dunham M H, "Data Mining: Introductory and Advanced Topics", Pearson Education, New Delhi, 2003.
T2	Jaiwei Han and Micheline Kamber, "Data Mining Concepts and Techniques", Elsevier,2006.
R1	Mehmed Kantardzic, "Data Mining Concepts, Methods and Algorithms", John Wiley and Sons, USA, 2003.
R2	Pang-Ning Tan and Michael Steinbach, "Introduction to Data Mining", Addison Wesley, 2006.

Online Resources

Sl.No	Website Link
1	https://www.coursera.org/specializations/data-mining
2	http://spoken-tutorial.org/